

# Neeraja Uppala

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## SUMMARY

Results-driven and meticulous Computer Science student with hands-on experience in full-stack development, cloud computing (AWS), and large-scale data processing. Proven ability to architect efficient, secure, and scalable solutions using Java, Python, Spring Boot, and React. Seeking a full-time Software Engineering role starting January 2026 to deliver high-impact results.

## EDUCATION

### North Carolina State University

*Bachelor of Science in Computer Science, Artificial Intelligence Concentration*

Raleigh, NC

*Aug. 2022 – Dec. 2025*

## EXPERIENCE

### Forensic Security Research Intern

*University College Dublin*

Dublin, Ireland

*May – July 2025*

- Architected and engineered a scalable Python framework to automate wordlist generation by interfacing with the DeepSeek LLM via the Ollama API, transforming a manual process into an automated pipeline
- Developed and optimized context-aware prompts through iterative testing to guide the LLM in generating targeted wordlists for password cracking applications, improving relevance and efficiency
- Benchmarked LLM-generated wordlists against traditional datasets (e.g., rockyou.txt) using HashCat and best64.rule; delivered comprehensive quantitative analysis on hit rate, complexity, and uniqueness to assess practical viability for digital forensic applications

### Software Developer Intern

*Ansys*

Austin, TX

*June – Aug. 2024*

- Achieved a 100x speedup in the TZR import workflow by optimizing object-oriented C++/Python code with advanced data structures and algorithmic techniques, enabling efficient processing of large-scale datasets
- Accelerated data processing further by parallelizing computational tasks using OpenMP, effectively distributing workloads across multiple threads to minimize processing time
- Recognized by senior leadership for delivering a critical performance enhancement that enabled faster data processing and improved system responsiveness for users

## PROJECTS

### Full-Stack Coffee Maker Management System | *Java, SQL, JavaScript, HTML/CSS*

- Developed a full-stack application following microservice principles using Spring Boot to create a secure REST API and connected it to a normalized MySQL database, implementing robust CRUD operations to ensure data integrity and efficient persistence for orders, inventory, and recipes
- Developed a dynamic and intuitive user interface using React and JavaScript that streamlined the order placement and fulfillment process, resulting in a seamless user experience
- Leveraged Postman for API endpoint integration testing, JUnit for backend unit testing, and Git for version control in an Agile development workflow to ensure code reliability and collaborative development

### AI Planning Agent for Chip's Challenge | *Java*

- Implemented an A\* search agent utilizing a Manhattan Distance heuristic and priority queue to enable real-time, optimal pathfinding within a complex, dynamic grid environment
- Designed a real-time state management system using object-oriented programming to track game elements (keys, chips, tiles), enabling adaptive decision-making in response to changing game states
- Significantly optimized computational performance, achieving a more than 50% reduction in pathfinding execution time while ensuring smooth and efficient agent navigation and action sequencing

### Gomoku/Renju Board Game | *C*

- Engineered a dual-mode command-line board game supporting both Gomoku and Renju variants, implementing distinct object-oriented gameplay rules and mechanics for each mode
- Architected a robust game state system with advanced move validation and board logic, ensuring strict adherence to variant-specific rules and flawless gameplay
- Enhanced user experience by developing a persistent save/load system via file I/O, allowing players to seamlessly interrupt and resume games

## TECHNICAL SKILLS

**Languages:** Python, Java, C++, C, C#, JavaScript, SQL, R, HTML/CSS

**Frameworks & Libraries:** React, Spring Boot, .NET, PyTorch, OpenMP

**Tools & Platforms:** Git, AWS, Jenkins, Postman, Docker, Jupyter, VS Code, Eclipse, Power BI, MySQL